

IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

R26-IT-134

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

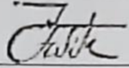
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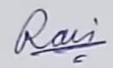
March 2026

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ABSTRACT

Visual impairment affects millions of people worldwide and often limits their ability to move safely and independently in daily environments. Traditional mobility aids such as white canes provide limited information about surrounding obstacles and cannot identify different object types. Although recent research has explored assistive technologies using sensors and computer vision, many existing systems rely only on distance sensors or general object detection models. These approaches may not provide sufficient contextual awareness for reliable indoor navigation.

This research focuses on developing an **AI-based object detection component** for a smart spectacle navigation system designed to assist visually impaired users. The proposed solution trains a lightweight object detection model capable of identifying common indoor obstacles such as chairs, doors, and stairs from images captured by a camera mounted on the spectacles. The model works together with IoT sensors that provide additional information about obstacle distance and user orientation.

The system uses technologies such as **machine learning, computer vision, edge computing, and IoT sensors**. The trained model is deployed on an edge device to perform real-time object detection and support safe navigation without relying on continuous internet connectivity.

The expected outcome of this research is a **lightweight and efficient object detection model** that improves obstacle awareness for visually impaired users. By combining visual detection with sensor information, the system aims to provide more reliable navigation assistance in indoor environments.

This research has both **social and commercial value**. Socially, it supports independent mobility and improves the quality of life for visually impaired individuals. Commercially, the developed model can be integrated into affordable assistive devices and smart wearable technologies.

Keywords: Visual Impairment, Smart Spectacles, Object Detection, Computer Vision, Assistive Technology

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor and co-supervisor for their valuable guidance, feedback, and support throughout the development of this research project titled *“IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired.”* Their insights greatly contributed to the success of this work.

I also thank the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment and resources required to conduct this research under the IT4010 Research Project module.

My appreciation extends to my group members for their collaboration and teamwork in developing the integrated assistive system.

Finally, I would like to thank my family and friends for their encouragement and support during the project.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AIoT	Artificial Intelligence of Things
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
FRS	Frames Per Second
FR	Functional Requirement
GPU	Graphics Processing Unit
GPS	Global Positioning System
IMU	Inertial Measurement Unit
IOT	Internet of Things
mAP	Mean Average Precision
ML	Machine Learning
NFR	Non-Functional Requirement
SDG	Sustainable Development Goal
SSD	Single Shot MultiBox Detector
UN	United Nations
WHO	World Health Organization
YOLO	You Only Look Once

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1. INTRODUCTION

1.1 Background and Context

Visual impairment is a major global issue that affects millions of people. “People who are blind or have low vision often face difficulties when moving around independently. Simple daily activities such as walking on the street, avoiding obstacles, or finding entrances can be challenging” [1]. Because of this, visually impaired individuals depend heavily on others or assistive tools to navigate safely.

According to the WHO, more than **2.2 billion people worldwide** have some form of vision impairment or blindness [2]. Many of them live in developing countries where access to advanced assistive technology is limited. “In countries like Sri Lanka, visually impaired individuals often rely mainly on **white canes or human assistance**, which do not always provide enough information about their surroundings.” [3]

Several technological solutions have been proposed to support visually impaired navigation. Some systems use ultrasonic sensors to detect obstacles and alert users through vibration or sound [4]. Other systems use camera-based object detection models such as YOLO to identify objects in the environment [5]. However, many of these systems have limitations. Vision-based systems may not perform well in low-light conditions, while sensor-based systems can detect distance but cannot recognize object types [6]. In addition, many existing systems do not effectively combine multiple sensors to improve accuracy [7].

To address these limitations, this project proposes a **smart spectacle system for visually impaired users**. The system combines different sensors and technologies to improve navigation safety. The device uses a **camera for object detection**, a **distance sensor to measure obstacles**, and an **IMU sensor to understand the user’s orientation**. A trained machine learning model processes this information and provides **audio feedback to guide the user safely**.

This project also supports the global development goals defined by the UN. It contributes to Sustainable Development Goal 3 by improving the quality of life of visually impaired individuals, and Sustainable Development Goal 10 by promoting inclusive technology that supports people with disabilities.

In terms of academic focus, this project aligns with the **Artificial Intelligence and Internet of Things (AIoT) research cluster**, as it integrates **machine learning, computer vision, and IoT sensors** to develop an assistive technology solution. The combination of these technologies aims to create a practical and affordable system that helps visually impaired people navigate their environment more safely and independently.

1.2 Literature Review

Recent research has explored different technological solutions to support navigation and independence for visually impaired individuals. Many studies focus on **wearable assistive devices**, especially smart glasses that combine artificial intelligence, sensors, and audio feedback to help users detect obstacles and understand their surroundings. [6]

Several studies use **computer vision and deep learning models** to recognize objects and provide guidance. For example, a smart guide glasses system based on YOLOv5 was developed to detect obstacles and provide real-time voice feedback to visually impaired users. The system achieved real-time detection performance with approximately **28 FPS and 74% detection accuracy**, demonstrating the potential of AI-based navigation support. [8]

Another study proposed an AI-based navigation system that uses a **camera and audio feedback** to identify obstacles and important landmarks in the environment. The system converts visual information into spoken instructions so that visually impaired users can move more safely and independently in different environments. [9]

Some researchers have focused on **IoT-based smart glasses** using sensors such as ultrasonic sensors and GPS modules. These systems detect obstacles and provide alerts through sound or vibration. For example, an IoT-enabled navigation system uses ultrasonic sensors and GPS to detect nearby objects and transmit location information for safer navigation. [4]

Other studies combine **AI and sensor technologies** to improve environmental awareness. For instance, a wearable navigation aid integrates a camera module with ultrasonic sensors and a deep learning model to detect objects and provide voice guidance in real time. [7]

Researchers have also explored **advanced wearable systems** such as smart glasses with object recognition, GPS navigation, and proximity sensors. These systems use neural networks to detect multiple objects and support navigation in both indoor and outdoor environments. [10]

More recent studies evaluate different **object detection algorithms** such as YOLO, SSD, and Faster R-CNN for indoor navigation assistance. These studies highlight the importance of selecting algorithms that balance **accuracy and real-time performance** for wearable devices. [5]

In addition to academic research, several commercial assistive products have been developed. Smart glasses and wearable vision devices provide functions such as text reading, object detection, and navigation assistance. These products use cameras and artificial intelligence to convert visual information into audio instructions for users. [11]

However, existing research and products still have several limitations. Many systems rely mainly on **camera-based object detection**, which may perform poorly in low-light environments or when objects are partially blocked. [6]

Sensor-based systems can detect distance but **cannot identify object types or environmental context** [4]. Additionally, many solutions do not effectively integrate multiple sensors, which limits the accuracy of obstacle detection and navigation guidance.

Therefore, further research is needed to develop **integrated smart assistive systems** that combine computer vision, sensor data, and intelligent processing to provide more reliable navigation assistance for visually impaired individuals.

Study / System	Technology Used	Key Features	Limitations
Wang et al. (2024)	YOLOv5 + Camera	Real-time obstacle detection with voice feedback	Primarily reliant on camera vision; performance degrades under poor lighting conditions
Visioneer AI Navigation	Camera + AI detection	Converts visual information into audio instructions	Requires high object detection accuracy for reliable guidance
Ultrasonic Sensor Navigation System	Ultrasonic sensors	Detects nearby obstacles and provides alerts	Cannot identify object types or provide environmental context
Wearable Navigation Aid	Camera + Sensors + AI	Combines object detection and distance sensing	Limited integration between sensor data and AI models
Commercial Devices (OrCam MyEye)	Camera + AI	Text reading, object recognition	High cost; limited dedicated navigation support

Table 1 Comparison of Existing Assistive Navigation Systems

1.3 Research Gap

Many existing assistive navigation systems designed for visually impaired individuals use object detection techniques, including models like YOLO, to recognize obstacles and provide audio-based guidance. However, these systems are often trained using general datasets such as the COCO dataset, which mainly include common objects like vehicles, animals, and household items. Such datasets are not specifically tailored to represent navigation-related hazards that visually impaired users frequently encounter, including stairs, narrow paths, poles, or low-hanging objects. Because of this mismatch, the models may fail to detect some of the most critical obstacles required for safe movement.

Another important issue is related to computational requirements. Many object detection models are developed and trained for high-performance computing environments. In contrast, wearable assistive devices such as smart spectacles operate on low-power edge hardware with

limited processing capability. This creates challenges when deploying large and complex deep learning models, as they may not perform efficiently without proper optimization.

To address these limitations, there is a need to develop more specialized model training approaches that focus on navigation-specific obstacles while also considering the constraints of edge devices. In this research, a lightweight object detection model is trained using a dataset focused on indoor navigation hazards. In addition, the model is designed to work efficiently on edge hardware and is further improved by incorporating contextual information from IoT sensors, enabling better obstacle awareness for visually impaired users.

2. OBJECTIVE

2.1 Main Objective

To design and develop a smart spectacle system that improves indoor navigation for visually impaired individuals by training a lightweight object detection model to identify navigation hazards and integrating it with IoT sensors to provide real-time audio guidance within the project development period.

2.2 Specific Objectives

1. To collect and prepare a labeled dataset of indoor navigation hazards such as stairs, doors, and obstacles for training the object detection model.
2. To train and optimize a lightweight object detection model capable of detecting navigation-related obstacles in real time.
3. To integrate the trained model with camera input and IoT sensors to improve obstacle awareness and navigation assistance.
4. To develop an audio feedback system that provides real-time alerts to visually impaired users based on detected obstacles.
5. To evaluate the performance of the trained model using detection accuracy and response time in indoor navigation scenarios.

3. METHODOLOGY

3.1 Research approach

This research is conducted under a research approach known as Design Science Research, where the main aim is to design and develop a technological artifact that addresses a real-world problem. The technological artifact that is being proposed in this research is a Smart Spectacle Hardware Module with Multi-Sensory Integration, which is aimed at helping visually impaired persons navigate indoor environments.

In addition, this research has adopted an experimental evaluation approach, where the effectiveness of the proposed system will be evaluated. The prototype that is being developed in this research will be used in controlled indoor environments to test its effectiveness in detecting obstacles and recognizing objects.

3.2 Technical methods and algorithms

This model uses a deep learning–based object detection approach to identify indoor obstacles that may affect visually impaired users. The system applies the YOLO algorithm to detect objects from images captured by the camera. YOLO is suitable for real-time applications because it processes images quickly while maintaining good detection accuracy. The trained model is designed to recognize navigation-related obstacles such as chairs, doors, walls, and stairs.

The model training process begins with **dataset preparation and image labeling**. Images of indoor environments containing navigation obstacles are collected to create a dataset. Each image is then annotated by marking the location of objects using bounding boxes and assigning the corresponding object class. This labeling process creates the training data required for the object detection model to learn how to identify obstacles within the environment. An example of the image annotation process is shown in Figure 1.

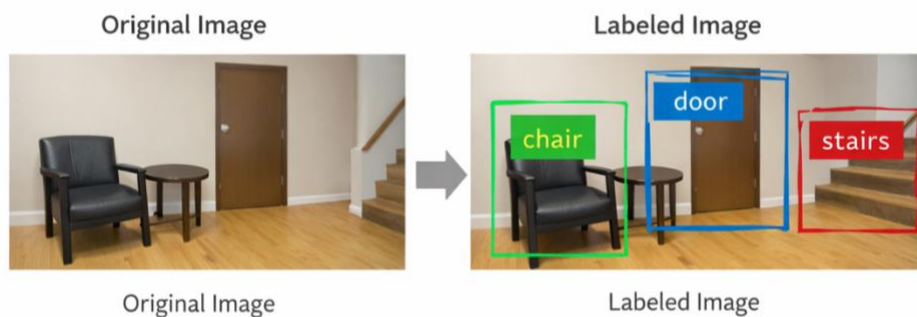


Figure 1 Image annotation process

The overall workflow of the model training process is illustrated in **Figure 2**.

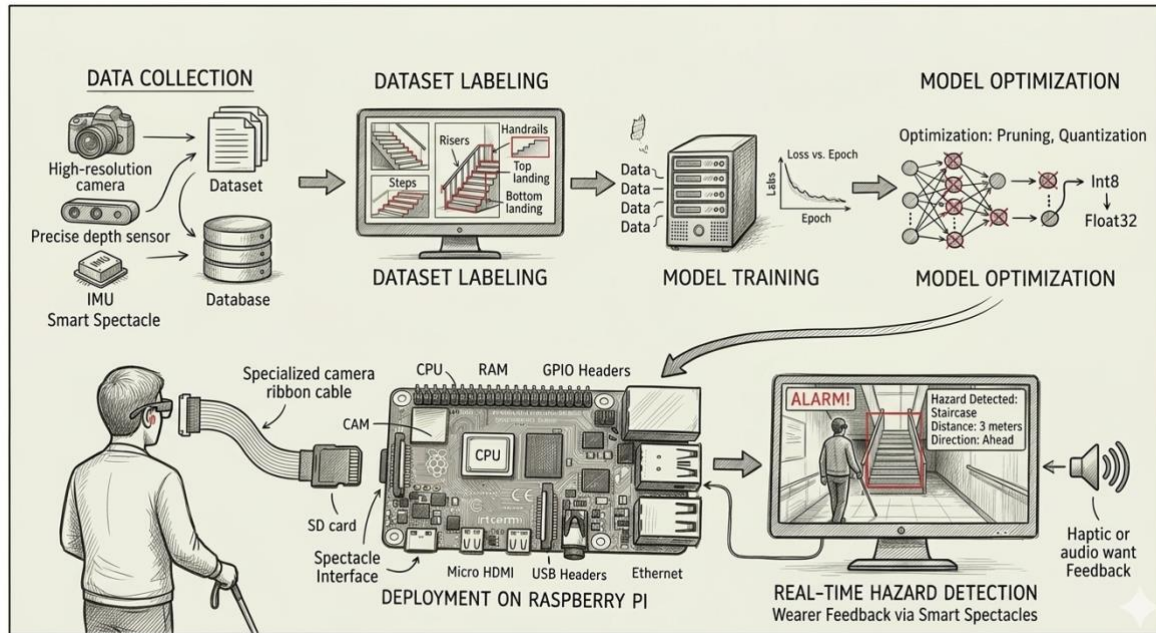


Figure 2 Object Detection Model Training Pipeline

As shown in Figure 2, the dataset is first collected and labeled before being used to train the YOLO-based object detection model. During training, the model learns patterns from the labeled images in order to accurately detect objects in new input images.

After the training process is completed, the trained model is optimized and deployed to an edge computing device used in the smart spectacle system. This allows the model to process camera input in real time and detect obstacles efficiently, enabling the system to provide navigation assistance for visually impaired users.

- **Object Detection Algorithm**

The proposed system uses an object detection approach to identify navigation hazards in indoor environments. A deep learning-based object detection model using YOLO is applied to detect obstacles from images captured by the camera. YOLO is selected because it can process images quickly while maintaining good detection accuracy, which is important for real-time applications. The trained model is designed to recognize common indoor obstacles such as chairs, doors, stairs, and walls that may affect the safe movement of visually impaired users.

- **Sensor Processing Algorithm**

In addition to camera-based object detection, the system integrates sensor data to improve environmental awareness. A LiDAR distance sensor is used to measure the distance between the user and nearby obstacles. This helps determine how far an object is from the user and

whether it represents a potential hazard. An IMU sensor is used to detect the orientation and movement of the user's head. This information helps the system understand the direction in which the user is facing and improves the accuracy of navigation assistance.

- **Hazard Detection Logic**

The system uses a simple decision logic to combine information from the camera and sensors. The object detection model first identifies the type of object present in the environment. The LiDAR sensor then measures the distance to the detected obstacle, while the IMU sensor provides information about the user's orientation. By combining these inputs, the system determines whether the detected object represents a navigation hazard. If a nearby obstacle is detected, the system generates an audio alert to warn the user and assist with safe navigation.

3.3 Tools and platforms with justification

Several hardware and software tools are used in this research to develop and train the object detection model and integrate it with the smart spectacle system.

Tool / Platform	Purpose	Justification
Python	Programming language for system development	Widely used for machine learning and computer vision with strong library support
OpenCV	Image processing and camera input handling	Efficient library for real-time image processing
YOLO	Object detection algorithm	Provides fast and accurate real-time object detection
TensorFlow Lite	Model deployment on edge devices	Optimized for running AI models on low-power hardware
Raspberry PI	Edge computing platform	Enables real-time processing and integration with sensors
LabelImg / Roboflow	Dataset annotation	Used to label images with bounding boxes for training the object detection model

Table 2 Tools and Platforms Used in the Proposed System

Table 2 summarizes the main tools and platforms used in the development of the proposed system. These tools support different stages of the project, including dataset preparation, model training, image processing, and system deployment. Python and OpenCV are used for implementing computer vision tasks and handling camera input, while the YOLO algorithm is applied to detect navigation hazards from captured images. TensorFlow Lite enables efficient deployment of the trained model on low-power devices, and the Raspberry Pi platform

provides edge computing capabilities for real-time processing. In addition, annotation tools such as LabelImg or Roboflow are used to label the dataset and prepare training data for the object detection model. Together, these tools provide a practical and efficient framework for developing the smart spectacle navigation system.

3.4 Data collection and ethical considerations

The data used in this research is primarily collected in the form of images representing common indoor navigation environments. These images include objects and obstacles that visually impaired users may encounter while moving indoors, such as chairs, doors, stairs, tables, and walls. The images are collected from indoor locations such as corridors, rooms, and staircases to ensure that the dataset represents realistic navigation scenarios.

After data collection, the images are prepared for training by performing **image annotation**. Each image is labeled by drawing bounding boxes around the relevant objects and assigning appropriate class labels. This labeled dataset is then used to train the object detection model so that it can learn to recognize different navigation hazards within indoor environments.

Ethical considerations are also taken into account during the data collection process. The dataset focuses only on environmental objects and obstacles rather than personal information. Any images collected during the experiments are used solely for research purposes, and care is taken to avoid capturing identifiable individuals. If people appear unintentionally in the images, their identity is not recorded or used in the dataset.

In addition, the system is designed to process data locally on the device without storing or sharing sensitive information. These measures ensure that the research follows responsible data usage practices while respecting privacy and ethical guidelines.

3.5 Validation strategy and performance metrics

The proposed system is validated through experimental testing to evaluate the performance of the trained object detection model and the effectiveness of the navigation assistance provided by the smart spectacle system. The evaluation is conducted in indoor environments such as corridors, rooms, and staircases where common obstacles are present. During testing, the camera captures real-time images and the trained model detects navigation hazards such as chairs, doors, stairs, and walls. The LiDAR sensor measures the distance to nearby objects while the IMU sensor provides head orientation information, enabling the system to generate appropriate navigation alerts.

To evaluate the performance of the trained model, several standard object detection metrics are used. **Precision** measures the proportion of correctly detected objects among all predicted detections, while **recall** measures the ability of the model to detect all relevant obstacles present in the environment. These metrics are calculated using the following equations:

Precision:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Where:

TP = True Positives (correctly detected objects)

FP = False Positives (incorrect detections)

FN = False Negatives (missed detections)

In addition to these metrics, **detection accuracy** is used to evaluate the overall performance of the object detection model.

The system is also evaluated based on real-time performance. **Response time** is measured to determine how quickly the system detects an obstacle and generates an audio alert for the user. **Distance measurement accuracy** evaluates how precisely the LiDAR sensor measures the distance to nearby obstacles. Finally, **navigation assistance reliability** is assessed by observing how effectively the system warns users about potential hazards while navigating indoor spaces.

The key performance metrics used in this research are summarized in Table 3.

Metric	Purpose
Precision	Measures the correctness of detected objects
Recall	Measures the ability to detect actual obstacles
Detection Accuracy	Evaluates overall object detection performance
Response Time	Measures the speed of hazard detection and alert generation
Distance Measurement Accuracy	Evaluates the accuracy of LiDAR distance detection

Table 3 Performance Metrics Used for System Evaluation

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of proposed solution and its components

The proposed solution is a smart spectacle-based assistive system developed to help visually impaired individuals navigate safely within indoor environments. It combines computer vision, IoT sensor technology, and edge computing to detect obstacles and provide real-time guidance.

The system is made up of several components that work together. A camera captures images of the user's surroundings, and these images are analyzed using an AI-based object detection model to recognize obstacles such as chairs, doors, and stairs. At the same time, IoT sensors, including distance sensors and an IMU, collect additional data such as how far obstacles are and the user's orientation. All this information is processed on an edge device like a Raspberry Pi, which then generates audio alerts to inform the user about nearby hazards, as illustrated in Figure 3.

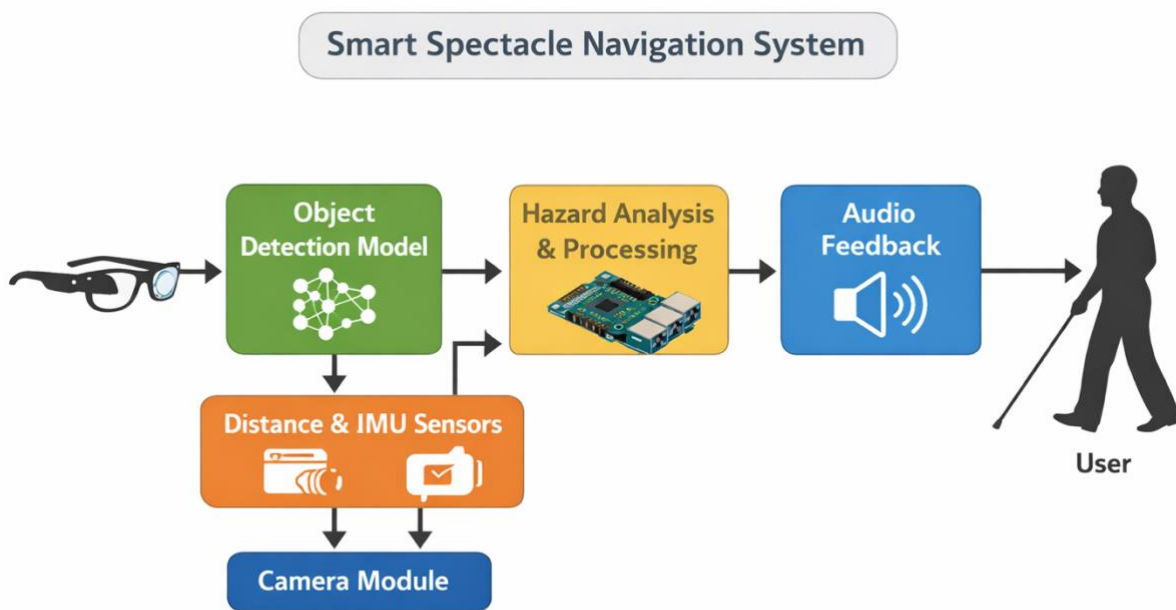


Figure 3 Overview of the Smart Spectacle Navigation System

4.2 System diagram of components

The smart spectacle navigation system consists of several interconnected components that work together to assist visually impaired users during indoor navigation. The system combines computer vision, IoT sensors, and edge computing to detect obstacles and provide navigation guidance.

The **camera module** captures images of the surrounding environment, which are processed by the **AI-based object detection model** to identify potential obstacles. This object detection model represents the core component of this research, as it is responsible for recognizing navigation hazards such as chairs, doors, and stairs from the captured images.

In addition to visual data, the system receives contextual information from **IoT sensors**, including distance sensors and an **IMU sensor**, which provide information about obstacle proximity and user orientation. These inputs are processed by an **edge computing device (Raspberry Pi)** that performs real-time analysis. Based on the processed information, the system generates **audio feedback** to alert the user about nearby obstacles.

The overall system architecture and the interaction between these components are illustrated in **Figure 4**, where the **object detection model component** is **highlighted as the main focus of this research**.

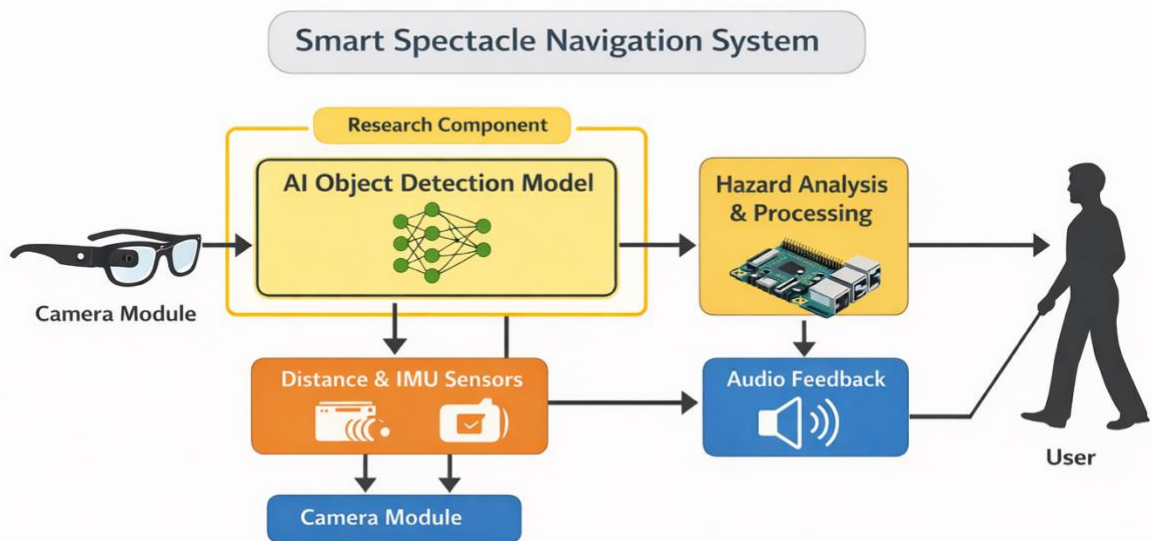


Figure 4 System Architecture of the Smart Spectacle Navigation System Highlighting the AI Object Detection Component

4.3 Highlight the individual component and its parts

The primary component developed in this research is the **AI-based object detection model**, which is responsible for identifying navigation hazards from images captured by the camera module. This component focuses on training and deploying a machine learning model capable of detecting indoor obstacles relevant to visually impaired users.

The development of this component begins with **dataset preparation**. Images representing indoor environments and common obstacles such as chairs, doors, stairs, and walls are collected. These images are then annotated using bounding boxes and class labels to create a labeled dataset that can be used for training the object detection model.

After preparing the dataset, the next step is **model training**. A deep learning-based object detection algorithm is trained using the labeled dataset so that the model can learn to

recognize different objects present in indoor environments. During this stage, the model adjusts its parameters through multiple training iterations to improve its detection accuracy.

Once the model has been trained, it is used for **inference**, which is the process of applying the trained model to new input images captured by the camera. During inference, the model analyzes each image frame and predicts the location and class of detected objects. The detected obstacles are then passed to the system for further processing, enabling the navigation system to generate appropriate alerts for the user.

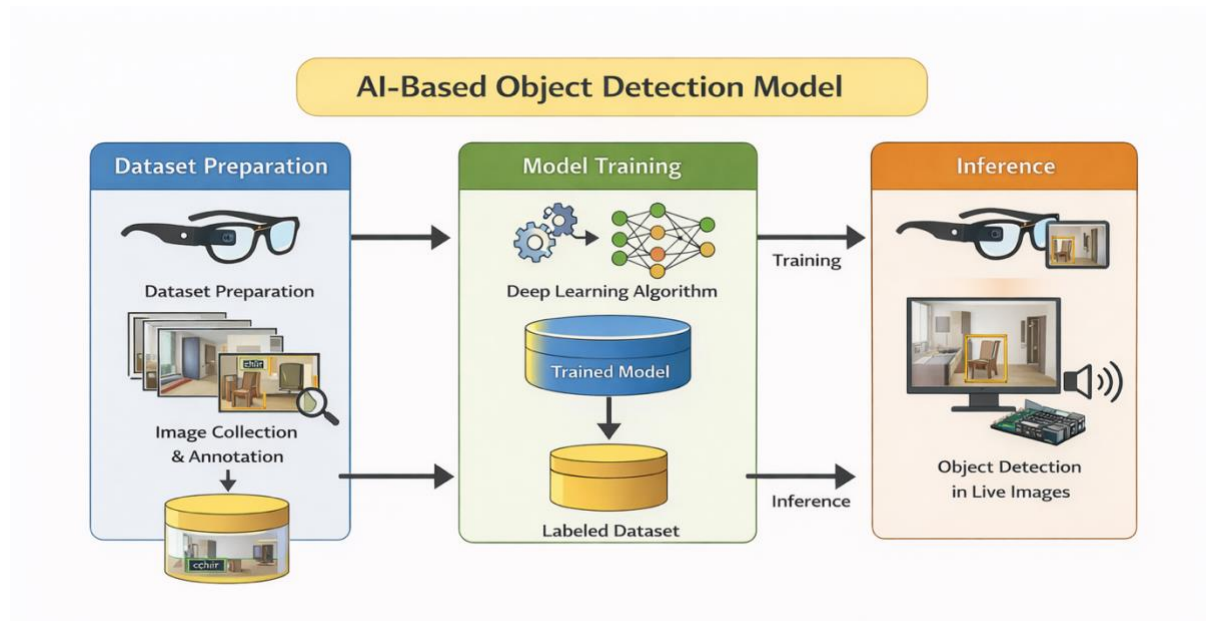


Figure 5 workflow of the AI based object detection model component

4.4 Component interactions

The interaction between system components occurs through the following steps:

- The **camera module** captures real-time images of the surrounding environment.
- The captured images are sent to the **AI object detection model**, which analyzes them to detect obstacles such as chairs, doors, or stairs.
- **IoT sensors**, including distance sensors and the **IMU sensor**, collect additional data about obstacle distance and user orientation.
- The outputs from the **AI model and sensor modules** are transmitted to the **edge processing unit (Raspberry Pi)**.
- The processing unit combines the visual detection results with sensor data to identify potential navigation hazards.

- Once a hazard is detected, the system sends the information to the **audio feedback module**.
- The audio module generates **voice or sound alerts** to guide the user safely during navigation.

4.5 Data flow explanation

- The process begins with the **camera module**, which captures images of the surrounding environment in real time.
- The captured images are transmitted to the **AI-based object detection model** for analysis.
- The model processes each image frame and identifies potential obstacles by predicting object classes and their locations.
- At the same time, **IoT sensors** such as distance sensors and the **IMU sensor** collect additional environmental data, including obstacle distance and user orientation.
- The outputs from the **object detection model and sensor modules** are sent to the **edge processing unit (Raspberry Pi)**.
- The processing unit combines the visual detection results with sensor information to determine whether a detected object represents a navigation hazard.
- Once a hazard is identified, the system generates an alert through the **audio feedback module**.
- The user receives the audio notification, allowing them to respond appropriately and navigate more safely.

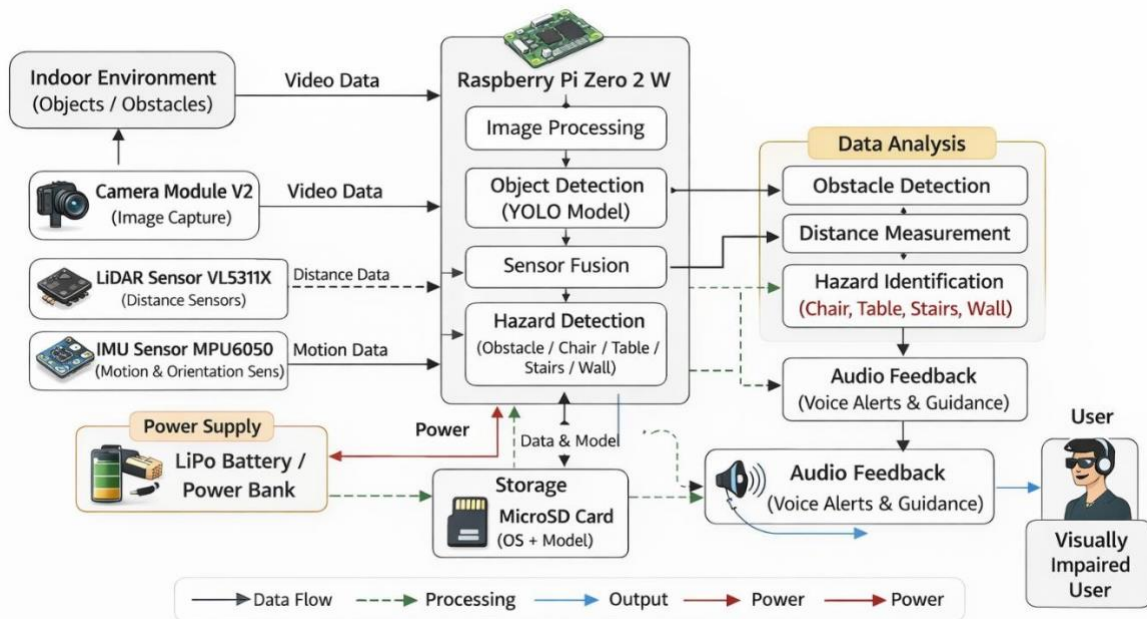


Figure 6 Data flow of the proposed system

4.6 Scalability and security considerations

Scalability

- The **object detection model** can be retrained with additional datasets to recognize new objects or navigation hazards in the future.
- The system architecture allows the integration of **additional sensors**, such as improved distance sensors or environmental sensors, to enhance obstacle detection accuracy.
- The model can be further optimized or connected to **cloud-based services** for advanced training, model updates, or dataset management if higher computational resources are required.

Security

- All visual data captured by the camera is **processed locally on the edge device**, reducing the need to transmit sensitive information over external networks.
- The system is designed **not to store personal or identifiable images**, ensuring that captured data is used only for real-time object detection.
- These measures help protect **user privacy and data security**, which is particularly important when developing assistive technologies for visually impaired individuals.

5. USER REQUIREMENTS

5.1 Stakeholder identification

The proposed smart spectacle navigation system involves several stakeholders who will interact with or benefit from the system.

- **Visually Impaired Users** – The primary users of the system who will rely on the smart spectacles for safe indoor navigation and obstacle detection.
- **Caregivers and Family Members** – Individuals who support visually impaired users and benefit from technologies that improve user independence and safety.
- **Researchers and Developers** – Individuals involved in designing, training, and improving the object detection model and system functionality.
- **Assistive Technology Organizations** – Organizations that promote technologies to improve accessibility for people with disabilities.

5.2 Functional requirements

The functional requirements describe the main functions that the system must perform.

- The system shall collect and prepare image data representing indoor navigation obstacles.
- The system shall label collected images using bounding boxes and class labels to create a training dataset.
- The system shall train an object detection model using the labeled dataset.
- The system shall detect navigation-related objects from camera images using the trained model.
- The system shall process input images and identify obstacles such as chairs, doors, and stairs in real time.
- The system shall output detected object classes and bounding box locations for further processing by the navigation system.
- The system shall allow the trained model to be deployed on an edge device for real-time inference.

5.3 Non-functional requirements

- **Performance**
The trained object detection model shall process input images and detect obstacles in real time with minimal processing delay.
- **Accuracy**
The trained model shall achieve a high level of detection accuracy when identifying navigation-related obstacles in indoor environments.
- **Efficiency**
The model shall be lightweight and optimized to run efficiently on edge computing devices with limited processing power and memory.
- **Reliability**
The object detection system shall consistently detect obstacles during repeated testing under different indoor conditions.
- **Scalability**
The model should support future improvements such as adding new object classes or expanding the dataset for better detection performance.
- **Security and Privacy**
The system shall process images locally and avoid storing or transmitting personal or identifiable information collected during operation.
- **Usability**
The trained model should integrate smoothly with the smart spectacle system to provide fast and reliable obstacle detection for navigation assistance.

- **Model Latency**
The object detection model shall perform inference within an acceptable time limit (e.g., less than one second) to support real-time navigation assistance.
- **Fault Tolerance**
The system should continue operating even if partial sensor data is temporarily unavailable.
- **Maintainability**
The system should allow future improvements such as retraining the model with additional datasets or updating model parameters without major system changes.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target users of the proposed smart spectacle system are visually impaired individuals who need assistive technology to move and navigate independently. Based on global statistics, a large number of people are affected by partial or complete vision loss, which creates a strong demand for affordable and effective assistive solutions.

The initial focus of commercialization is on visually impaired users living in urban areas, including students and working professionals who regularly move through indoor environments such as universities, offices, hospitals, and other public spaces. In addition to individual users, the system is also suitable for organizations such as rehabilitation centers, assistive technology providers, and educational institutions that support visually impaired individuals. These organizations can use the system to improve mobility training and enhance user independence.

A typical customer persona for this product can be described as follows:

Persona: “Independent Navigator”

- Age: 18–45
- Profile: Visually impaired student or professional
- Needs: Safe and independent movement in indoor environments
- Challenges: Difficulty identifying nearby obstacles such as stairs, doors, or furniture using traditional mobility aids
- Goal: Increase independence and confidence when navigating unfamiliar indoor spaces

By focusing on this user segment, the product addresses a **practical and everyday challenge faced by visually impaired individuals**, enabling them to move more safely without relying heavily on external assistance.

6.2 Value Proposition

The proposed smart spectacle system offers a **practical and intelligent assistive navigation solution** that combines computer vision and sensor technologies to detect obstacles in real time. Unlike traditional mobility aids such as white canes, which provide limited environmental information, the system uses an AI-based object detection model to recognize navigation hazards and provide immediate feedback to the user.

The key value proposition of the system lies in its ability to provide **real-time obstacle awareness through an affordable wearable device**. By integrating a lightweight object detection model with edge computing hardware, the system can operate without continuous internet connectivity, making it suitable for everyday use.

The main value offered by the system includes:

- **Enhanced Independence:** Enables visually impaired users to navigate indoor environments more confidently.
- **Real-Time Hazard Detection:** Uses AI to identify obstacles such as chairs, doors, and stairs.
- **Affordable Assistive Technology:** Designed to be more accessible compared to expensive commercial smart glasses.
- **Edge-Based Processing:** Processes data locally, ensuring faster response times and improved privacy.

Through these benefits, the proposed system aims to deliver a **cost-effective and practical assistive technology that improves mobility, safety, and independence for visually impaired individuals**.

6.3 Revenue Model

The primary revenue model for this component focuses on the **AI-powered object detection software used within the smart spectacle navigation system**. The trained object detection model can be commercialized as part of a complete assistive navigation device developed by the overall project team. Revenue can be generated through the **sale of smart spectacle devices that incorporate the trained model**.

In addition, the AI model can be offered as a **software component that can be integrated into other assistive technologies**, such as wearable navigation devices or smart mobility aids. Future revenue opportunities may include **software licensing, AI model updates, and**

system maintenance services provided to assistive technology providers or research institutions working with visually impaired users.

6.4 Cost Estimation

The cost related to this component primarily involves **AI model development, dataset preparation, and edge deployment optimization**. These costs mainly include computational resources used for training the model and development time required for data labeling, training, and testing.

However, when integrated into the complete smart spectacle system, the device will also require hardware components provided by the IoT module. An estimated hardware cost for the integrated device is shown below. (Table 4)

Component	Quantity	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	1	8,890
Raspberry Pi Camera Module V2	1	7,950
VL53L1X LiDAR Sensor	1	1,550
MPU6050 IMU Sensor	1	680
Li-Po Battery (3.7V 3000mAh)	1	1,550
MT3608 Boost Converter	1	150
MicroSD Card (32GB)	1	3,100
Earphone	1	810
Sunglass Frame and Wiring	1	1,000
Estimated Prototype Cost		~ 25,680 LKR

Table 4 Estimated hardware cost breakdown table

Estimated total system hardware cost:

Approximately **LKR 25,680 – 30,000 per unit**.

While these components belong to the IoT system module, the trained AI model represents the **core intelligence of the navigation system**.

6.5 Pricing Strategy

The pricing strategy aims to provide an **affordable assistive navigation solution for visually impaired users**, especially in developing regions such as Sri Lanka where many advanced assistive devices are financially inaccessible.

Considering the estimated hardware cost and software development effort, the integrated smart spectacle device could be priced in the range of **LKR 50,000 – 75,000 per unit**. This pricing allows for sustainable production while remaining significantly lower than many commercial assistive smart glasses available internationally.

Additionally, the AI model developed in this component could be provided to **assistive technology developers or device manufacturers as a licensed software module**, allowing further commercialization through software integration into other accessibility products.

6.6 Competitive Advantage

The key competitive advantage of this component lies in the **AI-based object detection model designed specifically for navigation hazards faced by visually impaired users**. Unlike general object detection systems trained on common datasets, this model focuses on detecting obstacles relevant to safe indoor navigation.

Key advantages include:

Specialized Model Training

The model is trained using datasets focused on indoor navigation hazards such as chairs, doors, and stairs, improving its relevance for assistive navigation applications.

Edge Device Optimization

The trained model is optimized for deployment on edge devices, allowing real-time obstacle detection without requiring continuous internet connectivity.

Low-Cost Implementation

The model can operate using affordable embedded hardware, making the overall system more accessible to users compared to many existing commercial solutions.

Expandable Architecture

The object detection model can be retrained with additional datasets to support new object categories and improve system performance over time.

6.7 Intellectual Property Considerations

The AI model training process and dataset preparation used in this component represent intellectual contributions that can be considered for protection under intellectual property frameworks.

The trained object detection model, training pipeline, and dataset annotations can be protected through **software copyright and research documentation ownership**. These elements represent the core intelligence of the navigation system and may form the basis for future development of assistive technologies.

Additionally, if the integration of computer vision models with wearable assistive systems introduces a novel technical solution, the overall system architecture could potentially be considered for **patent protection**. Protecting these intellectual assets can support future commercialization and encourage further development of AI-based accessibility solutions.

7. BUDGET AND JUSTIFICATION

This section outlines the estimated costs required to support the development and testing of the **object detection model used in the proposed navigation system**. The budget mainly covers hardware components required for model testing, data storage, and optional cloud computing resources used during the training process.

7.1 Hardware Costs

Component	Quantity	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	1	8,890
Raspberry Pi Camera Module V2	1	7,950
VL53L1X LiDAR Sensor	1	1,550
MPU6050 IMU Sensor	1	680
Li-Po Battery (3.7V 3000mAh)	1	1,550
MT3608 Boost Converter	1	150
MicroSD Card (32GB)	1	3,100
Earphone	1	810
Sunglass Frame and Wiring	1	1,000
Estimated Prototype Cost		~ 25,680 LKR

Table 5 Hardware costs breakdown table for Model training

Total Estimated Hardware Cost: LKR 25,680

Justification:

The Raspberry Pi device is used to deploy and test the trained object detection model in an edge computing environment. The camera module captures images that are processed by the trained model to detect navigation hazards. The storage device is used to store datasets, trained models, and experiment results during development.

7.2 Software Licenses

Software	Cost (LKR)
Python Programming Language	Free
TensorFlow / PyTorch	Free
OpenCV	Free
Annotation Tools (Labellmg / Roboflow basic version)	Free

Table 6 Cost breakdwon Table for Software licenses

Total Software Cost: LKR 0

Justification:

All software tools used in this research are **open-source technologies**, which eliminates licensing costs. These tools provide essential capabilities for dataset preparation, computer vision processing, and training of deep learning models.

7.3 Cloud Services

Service	Estimated Cost (LKR)
Cloud Dataset Backup Storage	1,000

Table 7 Cost breakdwon Table for Cloud services

Estimated Cloud Cost: LKR 1,000

Justification:

Cloud storage can also be used to securely store datasets and trained models during development.

7.4 Budget Summary

Category	Estimated Cost (LKR)
Hardware	25,680
Software	0
Cloud Services	1,000

Table 8 Cost breakdwon Summary

Total Estimated Budget: Approximately LKR 26,680

Additional Note:

The development and training of the object detection model are performed using an existing Mac laptop owned by the developer, therefore no additional cost is required for development hardware.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task breakdown per member

Member	System Component	Key Tasks
Member 1 (ME)	Intelligent Vision-Based Danger Detector 	Task 1: Collect and prepare image dataset for indoor hazards. Task 2: Configure YOLO object detection model for edge devices. Task 3: Train and optimize the model using labeled datasets. Task 4: Evaluate detection accuracy and performance. Task 5: Integrate detection system with the smart spectacle platform.
Member 2	Intelligent Security & Remote Guardian System 	Task 1: Design and develop the guardian mobile application. Task 2: Implement fall detection and inactivity monitoring algorithms. Task 3: Develop geo-fencing functionality for safe zones. Task 4: Integrate cloud database for real-time monitoring. Task 5: Implement SOS emergency alert system.

Member 3	Smart Spectacle Hardware Module with Multi-Sensory Integration 	Task 1: Design the wearable smart spectacle hardware structure. Task 2: Integrate Raspberry Pi with camera, LiDAR, and IMU sensors. Task 3: Implement sensor data acquisition and synchronization. Task 4: Develop audio output interface for navigation alerts.
		Task 5: Implement push-button control for system interaction.
Member 4	Hybrid Indoor Positioning System with Offline Capability 	Task 1: Develop indoor localization algorithm using IMU sensors. Task 2: Implement vision-based landmark recognition. Task 3: Integrate dead-reckoning for indoor positioning. Task 4: Combine vision and IMU data for hybrid localization. Task 5: Integrate positioning system with navigation and hazard detection modules.

8.2 Timeline alignment

The development process of the proposed IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired will be carried out in an aligned manner with an appropriate timeline so that it can be achieved systematically during the entire research period. The project will be achieved through different key milestones that will be included in the research process.

The project will be aligned with the following timeline:

- **Phase 1 – Literature Review and Problem Analysis (Month 1):**

Thorough research will be performed based on the existing research works on assistive technologies, indoor navigation systems, computer vision models, and IoT-based wearable devices.

- **Phase 2 – System Architecture and Design (Month 2):**
Design of the entire system, including hardware as well as software components, shall be conducted.
- **Phase 3 – Hardware and Software Development (Month 3-4):**
The development of the individual components of the entire system will be performed.
- **Phase 4 – System Integration and Testing (Month 5):**
All the components of the entire system will be integrated and the testing of the entire system will be performed.
- **Phase 5 – Evaluation and Documentation (Month 6):**
Analysis of the entire system, completion of the research report, and completion of the project's presentation.

8.3 Balanced workload distribution

The project workload is distributed among the four members of the group based on the technical expertise of each member and the needs of the proposed project. Each member of the group is responsible for the development of a particular module of the project, including vision-based hazard detection, the remote guardian monitoring system, the smart spectacle hardware module, and the hybrid indoor positioning system. This ensures the development of all the major parts of the project efficiently, with all the members of the group participating equally.

Besides the development of the project's individual parts, some activities such as integration, testing, and report writing will be undertaken jointly by all the members of the group to ensure the equal participation of all members in the successful completion of the research project.

9. GANTT CHART

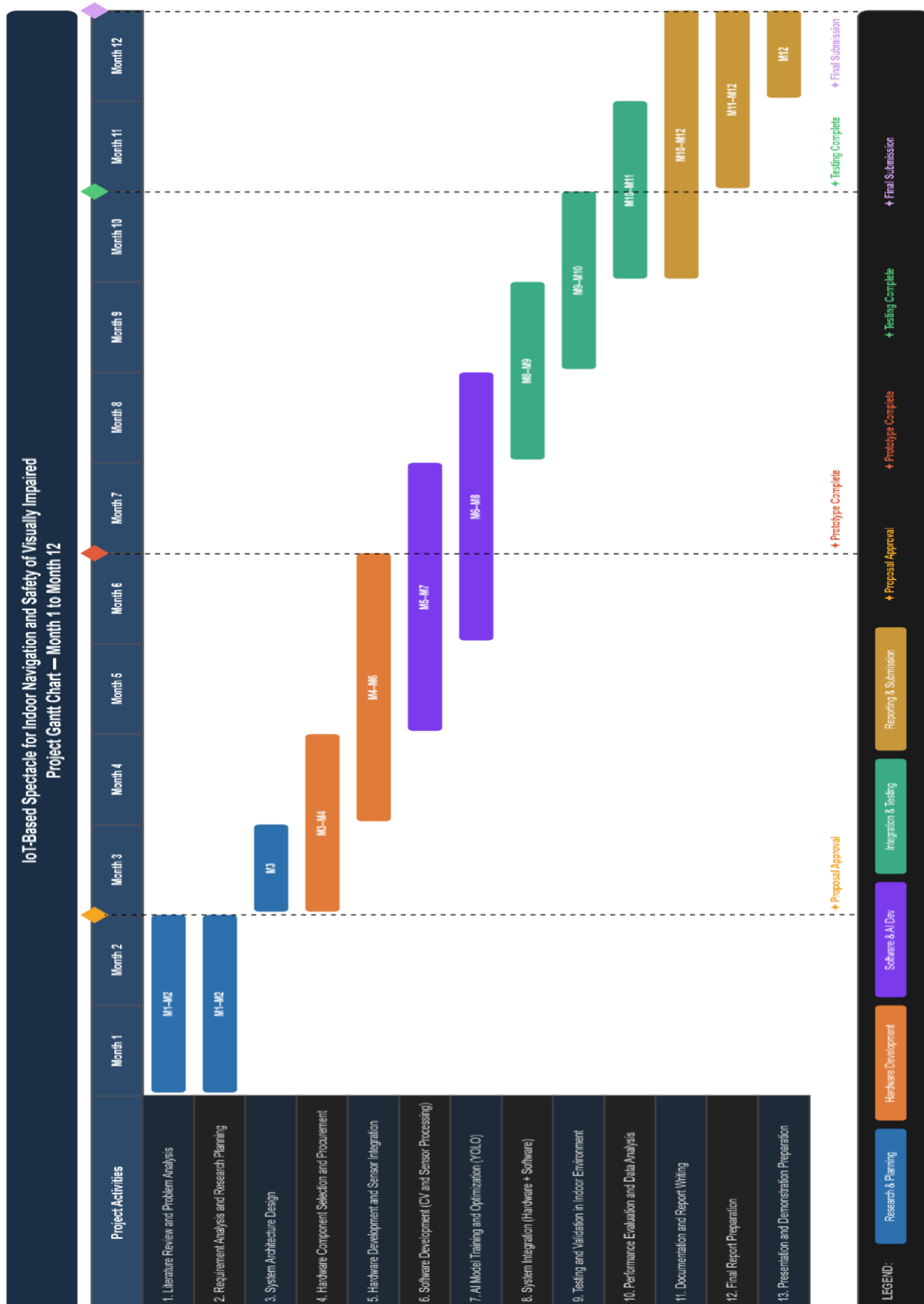


Figure 7 Gantt Chart

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11.APPENDICES

Appendix A - Survey Instruments

As a follow-up, the following is a sample questionnaire that seeks to understand the challenges faced by visually impaired individuals when moving around within an indoor environment. Additionally, the questionnaire seeks to find out the possible application of wearable assistive technology devices.

Target Audience: The target audience for the survey will be individuals who are visually impaired, caregivers of individuals who are visually impaired, and professionals who are knowledgeable about assistive technologies.

Survey Questions

1. What type of difficulties do visually impaired individuals commonly face when navigating indoor environments?
 - Difficulty identifying obstacles
 - Difficulty recognizing stairs or elevation changes
 - Difficulty determining direction or location
 - All of the above
2. Which assistive device is commonly used by visually impaired individuals for mobility?
 - White cane
 - Guide dog
 - Smartphone navigation applications
 - Wearable assistive devices
3. How important is obstacle detection for safe indoor navigation?
 - Very important
 - Important
 - Moderately important
 - Not important
4. What type of feedback would be most useful for a smart navigation system?
 - Audio instructions through earphones
 - Vibration alerts
 - Visual notifications
 - Combination of audio and vibration

5. How useful would a wearable smart spectacle device be for assisting visually impaired individuals in indoor navigation?
 - Very useful ○ Useful ○ Slightly useful ○ Not useful
6. Which feature would be most important in a smart navigation device for visually impaired users?
 - Obstacle detection ○ Staircase detection ○ Real-time navigation assistance ○ Emergency alert system

Appendix B - Detailed System Diagrams

• Smart spectacle hardware architecture diagram

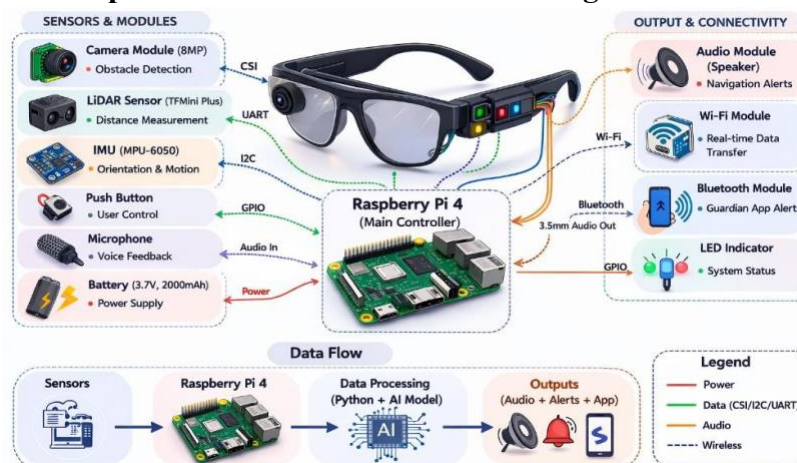


Figure 8 Smart spectacle hardware system diagram

• Data processing pipeline

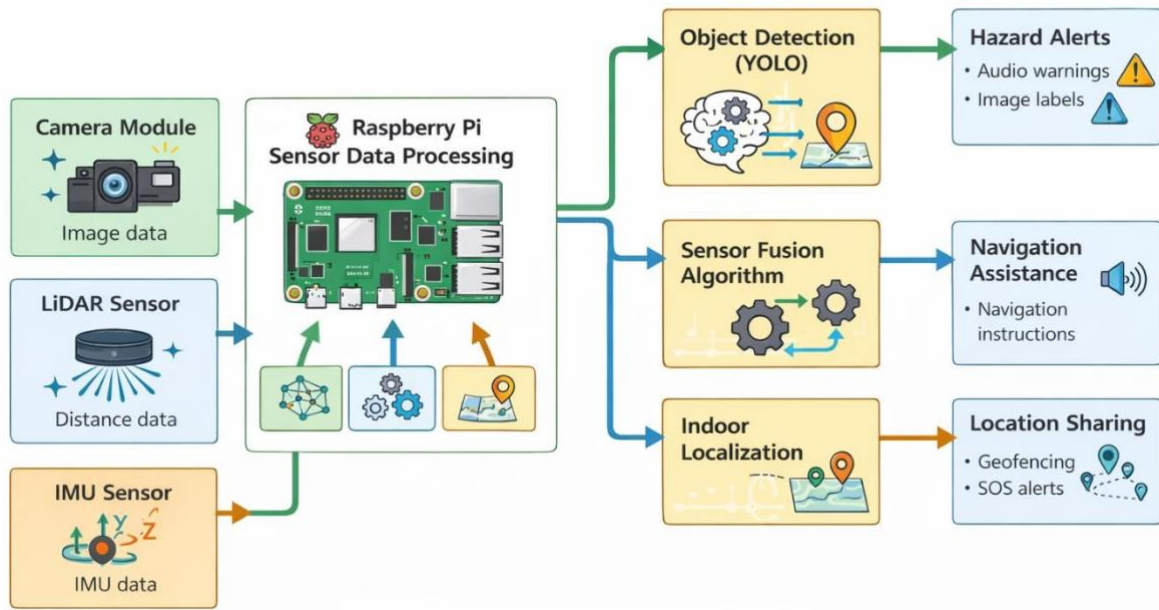


Figure 9 Sensor Data Processing Pipeline

Appendix C – Preliminary UI Mockups

This appendix presents the preliminary user interface designs for the Guardian Monitoring Mobile Application, which allows guardians to monitor the safety and location of visually impaired users using the proposed smart spectacle system.

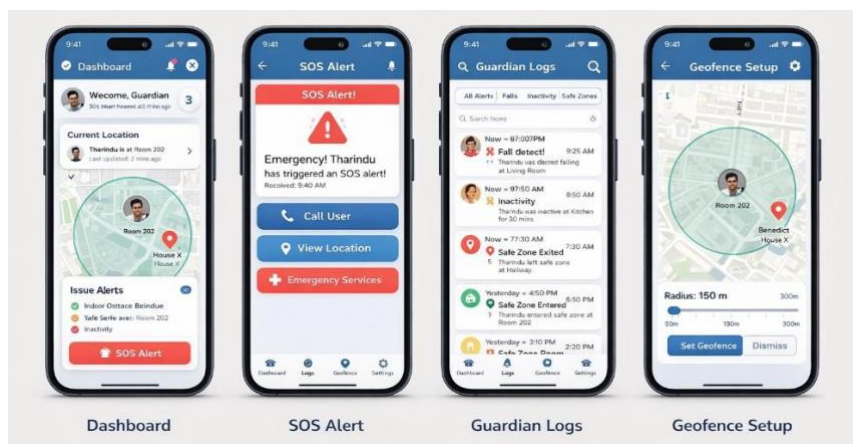


Figure 10 Preliminary User Interface Mockups for the Guardian Monitoring Mobile Application

Appendix D – Risk Analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Poor dataset quality or insufficient data	Low detection accuracy and poor model performance	Medium	Collect diverse indoor images and ensure proper annotation
Incorrect or inconsistent labeling	Model learns wrong patterns and reduces accuracy	Medium	Use annotation tools carefully and validate labels before training
Model overfitting	Model performs well on training data but poorly in real environments	Medium	Use data augmentation and validation dataset
Model too large for edge device	Cannot run efficiently on Raspberry Pi	High	Use lightweight models such as YOLOv8n and apply optimization (TensorFlow Lite)
Low detection accuracy for specific hazards (e.g., stairs)	Critical obstacles may not be detected properly	Medium	Add more targeted data for difficult classes
Slow inference speed	Delays in real-time obstacle detection	Medium	Optimize model size and reduce input resolution
Poor performance in low-light conditions	Reduced object detection accuracy	Medium	Include low-light images in dataset and combine with sensor data

Integration issues with hardware (camera input)	Model may not receive proper image data	Low	Test integration early and validate camera pipeline
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Table 9 Risk Analysis for Model Training

Appendix E – Ethical Considerations

This research is carried out following the guidelines on research ethics provided by the Sri Lanka Institute of Information Technology. This institute has an Ethics Review Committee (ERC), which deals with the review of the research proposal and ensures that all the research activities are carried out ethically.

The following ethical principles are considered in this research project:

- Personal Data:**
 This research does not require any personal data from any individual. If any data is required in the experiment for the system, it is related to objects in the environment, such as furniture, doors, staircase, etc.
- Privacy protection:**
 If images are captured during experiments in indoor environments, they will not intentionally include identifiable individuals. Any accidental appearance of people will not be stored or used for analysis.
- Responsible use of technology:**
 The developed smart spectacle system is only intended to assist and enhance the safety and mobility of people with visual impairments.
- Compliance with institutional ethics guidelines:**
 This research has been conducted based on the ethical guidelines suggested by the SLIIT ERC to carry out ethical academic research.
- Academic integrity:**
 All the information sources, data sets, and research material used in the current study are properly credited to ensure academic integrity.

SLIIT Ethics Review Committee:

<https://www.sliit.lk/about/about-sliit/ethics-review-committee>